Tweeting Fear: Analyzing Bidirectional Dynamics Between Trump's Posts and U.S. Market Volatility to Create Synthetic Tweets

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Research Question & Plan + Methods

Our research question is to determine whether a relationship exists between Donald Trump's social media posts (from his first precedence to current) and U.S market uncertainty. Specifically, we want to investigate: does a bidirectional relationship exist between the CBOE Volatility Index (VIX), Economic Policy Uncertainty Index (EPU), and the features (numerical and sentiment based) of Trump's tweets? If not, which direction has more predictive success? Regardless of the relationship found, we also aim to create a generative model that uses observed VIX/EPU values discretized into market volatility buckets to create synthetic Trump posts reflective of real market conditions.

In order to complete these research goals, we propose a hybrid unsupervised + supervised experimental pipeline that will ultimately allow us to attempt creating our VIX/EPU Trump Synthetic Tweet model. To begin, we pre-process Trump and VIX/EPU data into weeks from the beginning of Trump's first term to current. Before aggregating individual Trump tweets, we preprocess the tweet lexical content using two libraries/pre-trained models: VADER and FinBERT. We use VADER to assign numerical sentiment scores to individual tweets, then can later be aggregated to weekly statistics. FinBERT is used to convert tweets into single vector embeddings that capture finance-focused semantic relationships. Both of these transformed representations allow us to interact with the sentiment and semantics of the tweets.

After pre-processing, we begin supervised experimentation with baseline OLS in which we regress both a sub-set and all tweet features against same-week VIX values. We repeat his process with ridge regression and lasso regression in order to handle predictors that may be correlated. We then switch to unsupervised experimentation, in which we complete LDA topic modeling on the full tweet data set in order to discover high-level conceptual categories that his tweets fit into (like trade, MAGA, etc.). We also perform PCA + K-means on the FinBERT embeddings in order to see if we can find clusters for entire embedded tweets - we compare these clusters to our LDA results. After we have a better understanding of the latent space Trump's tweets 'come' from, we switch back to supervised learning. We perform non-linear experimentation using Random Forest and XGBoost; this will allow us to determine if relationships between volatility indices and Trump's tweets exist, but in non-linear spaces. We evaluate all models on temporal based test sets that are from time periods outside (after) the training data (test data from 2024-2026).

Lastly, the hallmark of our research and our most novel approach, we hope to use our learned tweet groupings along with twitter + VIX data to create a synthetic tweet model. We will take every tweet in our dataset, and label it with a discretized VIX value, discretized EPU value, topic (from unsupervised learning), and the tweet itself. We then pass all of these tweets to a GPT-2 model that we will train further. We will continue its training to predict the next word using our new dataset where the first words are the market and topic labels, followed by the tweet body. This will teach our model both how Trump writes, but what market conditions correspond with what he says and how he says it. This will allow us to create a model that ultimately creates synthetic Trump posts given just labels from processed VIX/EPU data. We are uncertain how well this will work, but are VERY excited to be working on it.

Data

We use the suggested Trump post [dataset] and [FRED-MD] data to access VIX & EPU values, aligned weekly (Saturday through Friday).

Summary Statistics

Table A: Trump Post Features — Weekly Aggregates (N = 208 weeks, Jan 2017 – Jan 2021)

Feature	Mean	Std Dev	Min	Max
Weekly Post Count	79.28	42.14	13.00	267.00
Mean VADER Compound	0.194	0.124	−0.105	0.567
Mean Retweets	19,065	5,453	9,898	53,453
Mean Post Length (chars)	170.24	33.92	69.73	234.51
Frac. All-Caps Words	0.064	0.028	0.016	0.205
Mean Exclamations / Post	0.807	0.168	0.222	1.537
Frac. Negative Posts (VADER < −0.05)	0.293	0.094	0.040	0.548

Table B: Volatility & Uncertainty Targets (weekly, Jan 2017 – Jan 2021)

Variable	Mean	Std Dev	Min	Max
VIX (weekly close)	17.877	9.247	9.140	66.04
ΔVIX (week-over-week, N=207)	0.048	3.849	−18.740	23.03
EPU Index (monthly, fwd-filled)	154.754	63.900	94.007	350.46

Baseline OLS Regression

Table C: Baseline OLS - Post Features vs Same-Week VIX Level | R^2: 0.332 | Adjusted R^2: 0.309 | Model $p < 0.001$ | N = 208

Feature	Coef.	Std Err	p-value	95% CI
Intercept	1.670	7.157	0.815	[−12.36, 15.70]
Weekly Post Count	0.060	0.015	< 0.001	[0.031, 0.089]
Mean VADER Compound	−8.437	12.333	0.494	[−32.61, 15.74]
Frac. Negative Posts	−30.277	15.986	0.058	[−61.61, 1.05]
Frac. All-Caps Words	37.096	31.588	0.240	[−24.82, 99.01]
Mean Retweets	0.001	0.000	0.010	[0.000, 0.001]
Mean Post Length (chars)	0.039	0.018	0.036	[0.003, 0.074]
Mean Exclamations / Post	4.916	4.703	0.296	[−4.30, 14.13]

We see that about 33.2% of all variation in VIX values can be explained by our tweet features, highlighting the need for further investigation.